DATABASE FOUNDATION & DESIGN

Core Concepts, Real-Life Examples & Practice

Learning Goal

Understand how a real-world system is converted into a database structure before moving to advanced SQL.

Major Topics

01	**Database Design Process**	How requirements become tables
02	**Entity Identification**	Finding important objects in a system
03	**Attributes**	Information belonging to each entity
04	**Keys**	Primary keys and foreign keys
05	**Constraints**	UNIQUE, NOT NULL, DEFAULT and CHECK
06	**Relationships & Cardinality**	1:1, 1:N and N:M
07	**Database Thinking**	Frontend, backend and database communication

1. Database Design Process

Database design means deciding what data an application needs, how the data should be organized, and how different pieces of data connect.

Simple Design Flow

Flow

Problem → Requirements → Entities → Attributes → Keys → Relationships → Constraints → ER Diagram → Tables

Real-Life Example: College Student System

A college wants software to manage students, departments, courses and attendance. Before creating tables, we understand what the college needs to store and how those things are related.

- Student information must be stored.
- Each student belongs to a department.
- Students can study multiple courses.
- Attendance must be recorded for students in courses.

Database Thinking

Start by asking: What data exists in this system and how is it connected?

2. Entity Identification

An entity is a real-world object or concept about which information needs to be stored.

System	Possible Entities
College	Student, Teacher, Department, Course, Attendance
Library	Member, Book, Author, Loan
Hospital	Patient, Doctor, Department, Appointment

Example

In a library, Book is an entity because information about books must be stored. Member is another entity with its own identity and information.

3. Attributes

Attributes are properties that describe an entity.

Entity	Attributes
Student	student_id, name, email, phone, date_of_birth
Book	book_id, title, isbn, category
Doctor	doctor_id, name, specialization, phone

Quick Distinction

Entity = what we store information about. Attribute = what information we store about it.

4. Keys

Primary Key: uniquely identifies each record in a table.

Foreign Key: connects one table to another.

Real-Life Example

Student.student_id can be the primary key. Attendance.student_id can be a foreign key pointing to Student.

5. Constraints

Constraint	Purpose	Example
PRIMARY KEY	Unique identity	student_id
FOREIGN KEY	Maintain relationship	department_id
UNIQUE	Prevent duplicates	email
NOT NULL	Value is required	student name
DEFAULT	Provide initial value	status = Pending
CHECK	Restrict valid values	attendance = P or A

6. Relationships & Cardinality

Relationships show how entities are connected. Cardinality tells how many records can participate in the relationship.

Relationship	Meaning	Real-Life Example
1 : 1	One record connects to one record	User ↔ User Profile
1 : N	One record connects to many records	Department → Students
N : M	Many records connect to many records	Students ↔ Courses

Important

A many-to-many relationship normally needs an intermediate table. Students and Courses can use Enrollment(student_id, course_id).

7. Database Thinking

A database developer should understand where the database sits inside a complete application.

Flow

Frontend → HTTP Request → Backend → Database Query → Database → Backend Response → Frontend

Example: Student Registration

A student fills a registration form. The frontend sends data to the backend. The backend validates the request and performs an INSERT operation. The database checks its rules, stores the student, and the backend returns a response.

Why Relationships Matter

• They prevent disconnected data.

• They allow retrieval of related information.

• They make JOIN operations meaningful.

• They represent real application rules.

Practice

Apply the concepts to a Library Management System.

Library Management System

Task	What to Produce
1	Identify at least 5 entities.
2	Give useful attributes to each entity.
3	Choose a primary key for every entity.
4	Identify foreign keys where needed.
5	Write relationships and cardinalities.
6	Add suitable constraints.
7	Explain how one library operation travels from frontend to database.

Quick Revision Questions

- What is an entity?
- What is the difference between an entity and an attribute?
- Why do we need a primary key?
- What does a foreign key do?
- Give one 1:N relationship from real life.
- How would you handle Students ↔ Courses N:M?
- Why use constraints even if the backend validates data?

Learning Check

You should be able to look at a simple real-world problem and identify its entities, attributes, keys, constraints and basic relationships.